

A Survey of Numerical Methods for Transient Eddy Current Problems

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Abstract -The purpose of this paper is to assess the efficiency of different time-stepping schemes and associated algebraic system solvers for transient eddy current computations.

NOTATION

In time-stepping schemes, the values of a (vector) quantity 'y' at three consecutive time steps are denoted by $y, \bar{y}, \bar{\bar{y}}$, respectively.

Primed symbols stand for trial (test) functions in the Galerkin method; time derivatives are denoted by dots or by ∂_t .

The scalar product $(u, v)_G$ of two vector functions u, v in a domain G is $(u, v)_G = \int_G u \cdot v \, dG$.

The matrix inequality $M > D$ for two symmetric matrices M, D means that their difference $(M-D)$ is positive definite.

A numerical solution $x(t) \in E^n$ is partitioned in a natural way, e.g. for the T- Ω method $x = (x_T, x_\Omega)$.

h - the time step, s - the maximum size of an element in the mesh.

ABBREVIATIONS

o.d.e.	ordinary differential equation
C-N	the Crank-Nicholson scheme
B-D	the Backward Difference (Euler) scheme
BDF	Backward Differentiation Formulae
R-K	Runge-Kutta method
DIRK	Diagonally Implicit R-K
SIRK	Singly Implicit R-K
SSI	Symmetrical Semi-Implicit scheme
EMP	Explicit scheme with a Matrix Parameter
CG	Conjugate Gradient method
ICCG	Incomplete Cholesky Conjugate Gradient method
QMD	Quotient Minimum Degree method
ND	Nested Dissection method
HM(IF)	Hybrid Method (with Incomplete Factorization)
HBM	Hierarchical Basis Multigrid method
MP	Multilevel Preconditioners

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I. INTRODUCTION

Applications of the Finite Element Method (FEM) to various formulations of the transient eddy current problem lead to ordinary differential-algebraic systems of equations. Most often these systems are time-stepped using C-N or B-D schemes, and the ICCG method [23] is applied at each time step as an algebraic solver. It is always very tempting, however, to avoid the solution of algebraic systems. Looking for possible alternatives to implicit schemes, we start with examining the matrix properties of the finite element formulations of the transient eddy current problems and then review the features of explicit and implicit methods. If implicit schemes have to be used, then obviously the most efficient algebraic system solvers should be employed. In the last part of the paper we compare different solvers for 2D and 3D problems: ICCG, HM(IF), ND, QMD, HBM and MP.

A generic transient eddy current problem which we are going to consider can be described as follows (Fig. 1): the computational domain G contains a subdomain with given currents $J_0(t)$ and a subdomain S ('screen') with unknown eddy currents $J(t)$. The problem may be two- or three-dimensional. For the sake of simplicity, we shall consider only linear problems with time-independent parameters and only simply connected regions in the 3D case. However, most of the analysis has relevance to nonlinear problems linearized by the Newton-Raphson method.

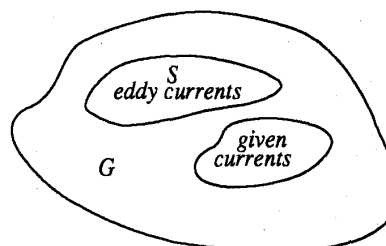


Fig. 1. A generic eddy current problem.

The eddy current problem is governed by Maxwell's equations with linear constitutive relationships $B = \mu H$; $J = \sigma E$:

$$\text{div } \mu H = 0; \quad \text{curl } E = -\partial_t(\mu H);$$

$$\text{curl } H = \sigma E \quad \text{in } S; \quad \text{curl } H = J_0 \quad \text{in } GS$$

As examples of finite element-Galerkin formulations, we shall consider the H- Ω , T- Ω or A^* - Ω methods in 3D and the integrodifferential formulation in 2D. All mentioned 3D methods utilize the scalar magnetic potential Ω outside the conducting domain S with the precomputed 'source field' H_0 satisfying $\text{curl } H_0 = J_0$.

In accordance with the objective of this paper, the following discussion of the formulations is concerned mainly with algebraic properties of the FEM matrices. Other features of these methods are studied in the references mentioned in the following sections.

II. FINITE ELEMENT FORMULATIONS

A. The integrodifferential formulation in 2D

If Fig. 1 represents a 2D configuration and the total current I through the cross-section of the conducting domain S is given, then the following formulation [19,29] for the one-component magnetic vector potential A and the gradient g of the electric scalar potential is valid:

$$\text{curl } v \text{ curl } A + \sigma(\partial_t A + g) = J_0$$

$$\int_S \sigma \partial_t A dS + g \int_S \sigma dS = -I$$

where $v = \mu^{-1}$ is the reluctivity. In the matrix (FEM-Galerkin) form this formulation is as follows [29]:

$$\begin{pmatrix} Q_A & 0 \\ L_{gA} & 0 \end{pmatrix} \frac{d}{dt} \begin{pmatrix} x_A \\ x_g \end{pmatrix} + \begin{pmatrix} L_{AA} & L_{Ag} \\ 0 & L_{gg} \end{pmatrix} \begin{pmatrix} x_A \\ x_g \end{pmatrix} = f(t)$$

with $L_{Ag} = L_{gA}$; $L_{AA} = L_{AA}^t > 0$; $Q_A = Q_A^t > 0$ and the diagonal matrix $L_{gg} > 0$.

B. The H- Ω method

The weak formulation directly in terms of the magnetic field H ($H = -\nabla \Omega$ in GS) is [6]:

$$(\sigma^{-1} \text{curl } H, \text{curl } H')_S + (\partial_t (\mu H), H')_G = -(\partial_t (\mu H_0), H')_G \quad (1)$$

where H, H' belong to a space of vector functions whose curl is square summable in G and equal zero in GS.

With a set of edge element basis functions chosen in S and a set of node element basis functions chosen in GS [6,32], the formulation (1) assumes the matrix form

$$Q \dot{x} + Lx = f; \quad Q = Q^t > 0; \quad L = L^t \geq 0 \quad (2)$$

where the matrix Q corresponds to the term $(\partial_t (\mu H), H')_G$ in (1), and the matrix L corresponds to the term $(\sigma^{-1} \text{curl } H, \text{curl } H')_S$.

C. T- Ω method

The vector potential of eddy currents T, $\text{curl } T = J$, is introduced in the conducting domain S [11,12,5,4,27]:

$$\text{div } \mu \nabla \Omega - \text{div } \mu T = \text{div } \mu H_0$$

$$\text{curl } \sigma^{-1} \text{curl } T + \partial_t \mu (T - \nabla \Omega) = -\partial_t \mu H_0 \quad \text{in GS}$$

The vector T has to be gauged by either imposing the non-divergence condition or reducing the number of components [11,12,1,18]. In any case the matrix form of the T- Ω formulation is

$$\begin{pmatrix} 0 & 0 \\ L_{T\Omega} & Q_T \end{pmatrix} \frac{d}{dt} \begin{pmatrix} x_\Omega \\ x_T \end{pmatrix} + \begin{pmatrix} L_{\Omega\Omega} & L_{\Omega T} \\ 0 & L_{TT} \end{pmatrix} \begin{pmatrix} x_\Omega \\ x_T \end{pmatrix} = f(t)$$

with $L_{\Omega\Omega} = L_{\Omega\Omega}^t > 0$; $L_{T\Omega} = L_{\Omega T}$; $L_{TT} = L_{TT}^t > 0$; $Q_T = Q_T^t > 0$.

D. A^* - Ω formulation

If the magnetic vector potential is used in the conducting domain S, and the scalar magnetic potential Ω is used in GS, then

$$\text{div } \mu \nabla \Omega = \text{div } \mu H_0 \quad \text{in GS} \quad (3a)$$

$$\text{curl } v \text{ curl } A + \sigma \partial_t A = 0 \quad \text{in S} \quad (3b)$$

The potentials A and Ω in (3) should be coupled on the interface boundary $\Gamma_{A\Omega}$. A suitable weak formulation is [14,26,27]:

$$\int_{\Gamma_{A\Omega}} \Omega \text{curl}_n A' d\Gamma + (v \text{curl } A, \text{curl } A')_G + (\sigma \partial_t A, A')_G = 0$$

$$(\mu \nabla \Omega, \nabla \Omega')_{GS} - \int_{\Gamma_{A\Omega}} \Omega' \text{curl}_n A d\Gamma = (\mu H_0, \nabla \Omega')_{GS}$$

The matrix form of this formulation is

$$\begin{pmatrix} 0 & 0 \\ 0 & Q_A \end{pmatrix} \frac{d}{dt} \begin{pmatrix} x_\Omega \\ x_A \end{pmatrix} + \begin{pmatrix} L_{\Omega\Omega} & -L_{\Omega A} \\ L_{A\Omega} & L_{AA} \end{pmatrix} \begin{pmatrix} x_\Omega \\ x_A \end{pmatrix} = f(t)$$

where $L_{\Omega\Omega} = L_{\Omega\Omega}^t > 0$; $L_{A\Omega} = L_{\Omega A}$; $L_{AA} = L_{AA}^t > 0$; $Q_A = Q_A^t > 0$.

E. Common features of the formulations

All the formulations outlined above, subject to certain constraints, assume the form (2), the matrix L being strictly positive definite for all methods except H- Ω .

For example, the A^* - Ω formulation, if considered (for the purposes of theoretical analysis, but not necessarily for actual numerical solution!) in the subspace

$$L_{\Omega\Omega} x_\Omega - L_{\Omega A} x_A = 0$$

takes on the form (2) with $Q = Q_A$, $L = L_{AA} + L_{A\Omega} L_{\Omega\Omega}^{-1} L_{\Omega A}$.

In the case of the T- Ω formulation, a similar subspace constraint is $L_{\Omega\Omega} x_\Omega + L_{\Omega T} x_T = 0$; then (2) holds with $Q = Q_T - L_{T\Omega} L_{\Omega\Omega}^{-1} L_{\Omega T}$, $L = L_{TT}$.

The behavior of the solutions to (2) is governed by the eigenvalue problem $Lx = -\lambda Qx$. It can be expected (although is not easy to prove rigorously) that the minimum and maximum eigenvalues $\lambda_{\max}, \lambda_{\min}$ are in the same order of magnitude as for the diffusion equation with the Laplace

operator: $|\lambda_{\min}| = O(1)$, i.e. does not depend on the mesh size s ; $|\lambda_{\max}| = O(s^{-2})$. Therefore the o.d.e. system (2) is stiff for all formulations.

III. TIME-STEPPING METHODS

It is well known that stability requirements are of great importance for stiff methods. Since the eigenvalues corresponding to (2) are real negative, A_0 (unconditional) stability [15,20] is relevant. A_0 -stability means that the numerical method is stable for arbitrary $h\lambda < 0$.

Explicit multistep, predictor-corrector or R-K methods cannot be unconditionally stable [15,20]. The stability regions of these methods are too small for stiff problems. For example, the intervals of absolute stability for p-stage Runge-Kutta methods on the real $h\lambda$ -axis are: $(-2,0)$ for $p=1$ or $p=2$; $(-2.51,0)$ for $p=3$; $(-2.78,0)$ for $p=4$.

Therefore a small time step must usually be chosen to ensure that $|h\lambda_{\max}|$ is less than the stability constant of the explicit scheme. Since $\lambda_{\max} = O(s^{-2})$, the maximum possible time step is $O(s^2)$. The finer the mesh, the smaller the time step that must be chosen. This conclusion is well known for diffusion-type problems.

Some non-standard schemes can be explicit and unconditionally stable at the same time. An example is given by the following scheme (EMP) for the o.d.e. (2) [28]:

$$M\bar{x} = (2M - Q - 1.5hL)\bar{x} - (M - Q - 0.5hL)\underline{x} + 1.5hf - 0.5h\underline{f}$$

where M is a symmetric positive definite matrix parameter. The EMP scheme has the first order of accuracy in time. The stability of the EMP scheme in the linear case is governed by the eigenvalues of the problem

$$\left[\lambda^2 M - \lambda(2M - Q - 1.5hL) + (M - Q - 0.5hL) \right] \bar{x} = 0$$

A sufficient condition for λ to lie in the unit circle is $2M > Q + hL$. In particular, M can be chosen as

$$M_i = \sum_j (|Q_{ij}| + 0.5gh|L_{ij}|), \quad g > 1, \quad i=1,2,\dots,n$$

It can be checked that for linear problems the EMP method coincides with the SSI scheme [22,17] if $Q=I$, $M=I+hD$. The stability condition is also satisfied in this case.

The SSI scheme, indeed, works reasonably well for the test problem considered in [22], which is a strongly nonlinear 1D heat conduction equation taken from [25, p.192]. However, this problem is only mildly stiff. For stiffer problems the EMP-SSI scheme fails to be efficient, as its accuracy deteriorates substantially. A series of numerical experiments [28] for the diffusion equation $\partial_t A - \nabla^2 A = J$ in a square region showed that for the EMP scheme the time step has to be taken 200-250 times smaller than for the C-N scheme in order to maintain the same accuracy. As a result, the EMP-SSI scheme required 100 times more computational time than the C-N method to obtain the numerical solution on the same time interval.

Thus, unfortunately, there seems to be no reasonable alternative for using implicit methods. Even apart from the stiffness of (2), the presence of the matrix coefficient Q of the derivative $\dot{x}$ by itself creates difficulties for explicit schemes.

Implicit schemes can be not only A_0 -stable, but L-stable [15,20]. L-stability ensures that very quickly decaying components of the solution are suppressed. If a method is not L-stable (e.g. the C-N scheme), then fast components of the theoretical solution may be represented by very slowly decaying components of the numerical solution. This is the reason for the commonly experienced 'noise' phenomenon when the C-N method is used [33].

A group of L-stable schemes is known as BDF [15,20]. The one-step scheme ($k=1$) of this group is the usual B-D (Euler) method. The two-step L-stable scheme is Gear's method: $\bar{x} - \frac{4}{3}\underline{x} + \frac{1}{3}\underline{x} = \frac{2}{3}hf$.

Among R-K methods, there exist L-stable *implicit* schemes of high orders. However, these systems are generally much more difficult to solve than systems for implicit linear multistep methods. Two families of implicit R-K methods - DIRK and SIRK [9,10] - make the solution more efficient, but only if *direct* (as opposed to iterative) solvers are used.

In the next section we review and assess the existing solvers that can be used in conjunction with the implicit schemes.

IV. SYSTEM SOLVERS FOR IMPLICIT SCHEMES

Traditionally several numerical methods have been used for the solution of FEM analogs of the eddy current equations. These methods for 2D problems were surveyed in [24].

Direct methods are based on the (complete) Cholesky factorization of the matrix. The simplest of the direct methods are band and envelope matrix factorization; however, they require $\theta = O(n^2)$ arithmetic operations and $C = O(n^{1.5})$ memory for 2D problems (n - dimension of the FEM matrix). The QMD and ND methods [16] are more efficient. For ND, $\theta = O(n^{1.5})$ and $C = O(n \log n)$. Theoretical estimates for QMD are not available, but practical experience indicates that QMD performs very similarly to ND, at least in the range $1 \leq n \leq 10^5$ (see section V). All of these direct methods are not effective for 3D problems: the computational time and memory increase to $\theta = O(n^2)$ and $C = O(n^{4/3})$.

Iterative methods seem to be more widely used because of low memory requirements: for most of them $C = O(n)$. The ICCG method is commonly used for preconditioning. HMIF [24] in 2D performs better than ICCG, but the difference is not significant enough to warrant switching to HMIF from ICCG. The HM [21], although slower than ICCG and HMIF, has an advantage of being robust: it works for all positive definite stiffness matrices, whereas methods based on incomplete factorization are theoretically justified only for M-matrices [23].

During the last ten years new classes of methods, potentially much more efficient than the traditional ones, have been

developed:

(a) *Domain Decomposition Methods* [7] are based on substructuring of the domain into subdomains. The preconditioner in [7] consists of the restriction of the operator to the subdomains and a specially constructed operator on the inner boundaries. For 2D problems the condition number of the preconditioned system is $O(d/s)$ [7], where d is the size of a subdomain. However, from the practical point of view, these methods are not easy to implement, especially in 3D.

(b) *multilevel (or multigrid) methods* ([30,8,3,31]) The finite element mesh has to be constructed as a sequence of nested meshes: the initial coarse mesh is refined by subdividing each triangle into two or four smaller triangles, and then this process is repeated recursively. HBM [30,3,31] and MP [8,31] methods have excellent theoretical characteristics: $\theta=O(n \log n)$, $C=O(n)$. For HBM this is only valid in 2D case (the rate of convergence deteriorates for 3D problems), whereas the MP method has the same rate of convergence for 2D and 3D problems. In addition, MP is exceptionally well suited for parallel computations.

V. NUMERICAL EXAMPLES

Numerical methods were tested with the TEAM Workshop Problem 11 - a hollow sphere in the step field. This benchmark problem is completely described in [13]. Only 2D solutions will be reported in this paper. The 2D formulation was based on the one-component magnetic vector potential in the cylindrical coordinate system. Although this problem is relatively simple, it does give a good idea of the performance of different solvers.

A sequence of six nested meshes has been generated, the coarsest one having 78 nodes and the finest one having 66449 nodes. Fig. 2, 3 show the theoretical solution [13] and the numerical solution obtained with the C-N and B-D scheme. In this example, the 'noise' associated with the C-N scheme is not significant even for relatively large time steps. However, the B-D scheme appears to be more accurate at the initial 'fast transient' stage of the process than the C-N scheme.

The computational time per time step (for SPARCTM station 2, FORTRAN compiler without optimization) and memory required for different solvers are shown in Fig. 4,5. (SPARCTM is a trademark of Sun Microsystems, Inc.). The initial condition for iterative methods was zero and iterations were stopped when the Euclidean norm of the residual was reduced by a factor of 10^{-5} .

The memory reported in Fig. 5 comprises all arrays needed for the implementation of methods (including the mesh data), assuming 4 bytes for integers and 8 bytes for real numbers.

VI. CONCLUSIONS

Some of the features of the numerical methods reviewed in this paper are summarized in the tables below.

For 2D problems direct methods ND and QMD are preferable to ICCG, provided enough memory is available. The MP method is highly efficient for magnetically homogeneous problems both in 2D and 3D. Efficiency of multilevel methods for inhomogeneous problems, although proven theoretically, needs further practical investigation.

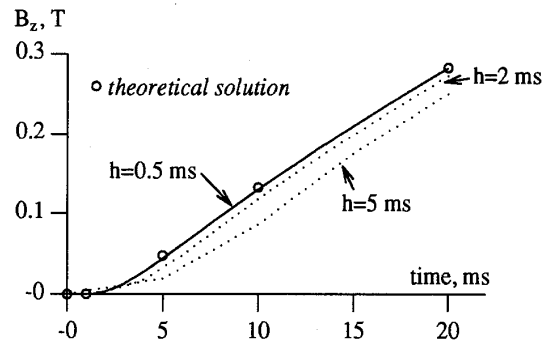


Fig. 2. Field B_z at the center of the sphere (C-N scheme); mesh: 16713 nodes, 33024 elements.

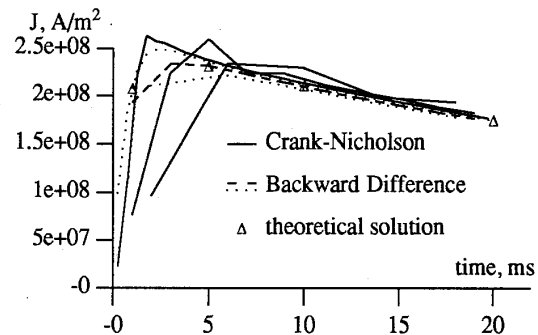


Fig. 3. Current density (A/m^2) at $y=5.25$ cm, $x=z=0$, $h=0.5$, 2 and 4 ms; mesh: 16713 nodes, 33024 elements.

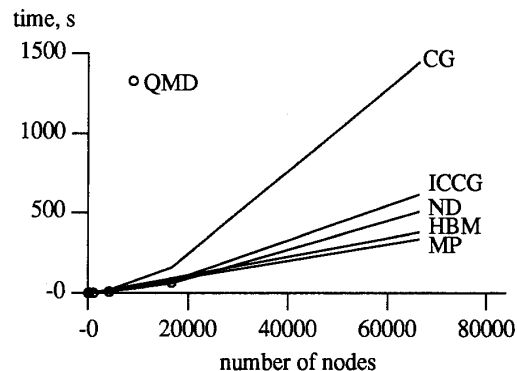


Fig. 4. Solution time for different methods

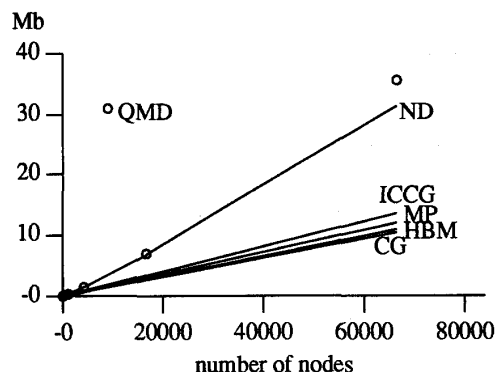


Fig. 5. Memory (in Mbytes) for different methods

TIME-STEPPING SCHEMES

METHOD	ADVANTAGES	DISADVANTAGES
B-D	L-stable	Implicit; first order
C-N	A_0 -stable second order	Implicit; not L-stable
BDF	L-stable; arbitrary order	Implicit; multistep
Explicit R-K	Explicit, simple	Small stability regions
DIRK, SIRK	Excellent stability	'Too' implicit Direct solvers required
EMP-SSI	Explicit and A_0 -stable	Low accuracy Not L-stable

CHARACTERISTICS OF SOME SOLVERS

Method	θ , 2D	C, 2D	θ , 3D	C, 3D
ICCG	$O(n^{1.5})$	$O(n)$	$O(n^{4/3})$	$O(n)$
HM(IF)	$O(n^{1.5})$	$O(n)$	$O(n^{4/3})$	$O(n^{4/3})$
QMD, ND	$O(n^{1.5})$	$O(n \log n)$	$O(n^2)$	$O(n^{4/3})$
HBM	$O(n \log n)$	$O(n)$	$O(n^{7/6})$	$O(n)$
MP	$O(n \log n)$	$O(n)$	$O(n \log n)$	$O(n)$

Note: theoretical estimates for QMD are not available

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In February 1976 Dr. Chari joined the Corporate Research and Development Electromechanics Branch, where he was involved in the electromagnetic analysis and design of rotating electrical machinery. In 1979 Dr. Chari was designated Manager, Electromagnetics Program. In 1992 he retired from the General Electric Company and still serves as a consultant to various programs at the Corporate Research and Development Center.

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